Assignment 3: We are social beings!

CGS616: Human Centered Computing

Due date: March 31st EOD

1 Background

In the previous assignments, you modeled **individual cognitive and behavioral states** using Markov processes derived from real-world observations. These included constructs such as:

- Sustained Attention, Vigilance
- Mood Swings, Coping
- Compliance
- Empathy, Trust

These models captured how an individual's internal states evolve over time.

In this assignment, we take a conceptual leap:

From individual cognition to ***social cognition and interaction***

We now ask:

> How do interactions between individuals give rise to **emergent societal patterns** such as cooperation, norm formation, or widespread violation?

Your previously modeled states (e.g., compliance vs violation) are no longer just internal transitions—they are now viewed as outcomes of **strategic interaction between agents**. Therefore, behavior such as compliance is not just an individual trait—it may emerge from repeated interactions, social influence, and network structure.

2 Objective

You will extend your Markov model into a multi-agent interaction framework using the Iterated Prisoner's Dilemma (IPD). More guidelines on the behavioral strategies you may use, how you may model evolution across generations, and how you may link them with your earlier assignments are provided in this document. You may use the base code discussed in the class for reference (refer to HelloIITK resource page).

In this assignment, you will:

- Propose a hypothesis
- Simulate agent interactions using fixed strategies
- Model repeated interactions (IPD)
- Introduce network structure
- Allow strategies to evolve
- Compare simulated transitions with real observed data

3 Interpreting Your Previous Model

You may map your previous states into the IPD framework. For example: one could think of categorizing the earlier identified individual cognitive states as C or D-

- (C): socially beneficial action
- (D): individually selfish action

And for this assignment, assume

- C $\rightarrow$ Cooperation
- D $\rightarrow$ Defection

Compute the state probabilities, and the transition matrix,

$$\begin{bmatrix} P(C \rightarrow C) & P(C \rightarrow D) \\ P(D \rightarrow C) & P(D \rightarrow D) \end{bmatrix}$$

may be interpreted as an emergent outcome of repeated interactions between agents.

4 Modeling Interaction: Who is the Other Player?

Who does an agent interact with?

You must explicitly choose and reason one of the following:

4.1 Social Norms Perspective

Agents interact with other agents (e.g., drivers, social media users)

Use this if your focus is:

- Social influence
- Emergence of cooperation or violation
- Peer effects (Captures reciprocity, retaliation, and norm formation)

4.2 Policy / Enforcement Perspective (Agent vs Regulatory Body)

The regulatory body may not "play" the game in the same way—it reshapes the payoff matrix in form of updates (e.g., through penalties, monitoring, incentives). So, you may choose to model this as updates in payoff matrix happening at the population level.

Use this if your focus is:

- Enforcement
- Policy interventions
- Incentive design

4.3 Combined Model (Bonus)

- Agents interact socially (IPD)
- The regulatory body dynamically modifies payoffs during iterations

5 Base Game Model

All agents play a Prisoner's Dilemma game.

$$\begin{array}{cc} C & D \\ C \quad (R,R) & (S,T) \\ D \quad (T,S) & (X,X) \end{array}$$

$$T > R > X > S$$

Example:

$$R = 3, \quad T = 5, \quad X = 1, \quad S = 0$$

6 Final Reporting: Needs a Hypothesis-Driven investigation

Submit a video presentation and the code in HelloIITK like the last time.

Note: You must select and test one clear hypothesis, and the entire reporting can be about whether your hypothesis explains the earlier observed state and transition probabilities

Examples include:

- "The network structure is key to widespread violation"
- "Compliance is not a fixed trait, but an emergent equilibrium"
- "Violation spreads through retaliation, not just selfishness"
- "Forgiveness mechanisms stabilize social systems"
- "Network structure determines whether norms survive or collapse"

Visualize: Strategy Frequency Tracking

Let:

$$x_k(t) = \text{fraction of strategy } k$$

Plot:

$$x_k(t) \text{ vs generation}$$

Visualize: Strategy Leaderboard at the start and end of evolution

$$\bar{P}_k = \frac{1}{N_k} \sum_{i \in k} P_i$$

6.1 Experimentation

You may choose experimental cases like the below to test the hypothesis:

One Bad Actor Experiment

Insert a single Always Defect (AD) agent in a Tit-for-Tat (TFT) population.

Forgiveness Threshold Study

Say, Compare TFT vs TF2T vs GTFT

Hub Effect in Networks

Place say an AD or AC agent at a central node in a scale-free network.

Noise vs Strict Punishment

Test Grim Trigger under noisy conditions (Policies).

7 Other useful information on Methods

7.1 Strategy Types

Each agent is assigned a fixed strategy.

Sample Strategies you may simulate

Always Cooperate (AC)

- Always plays C regardless of opponent behavior
- Vulnerable to exploitation

Always Defect (AD)

- Always plays D
- Maximizes short-term payoff but destroys cooperation

Tit-for-Tat (TFT)

- First move: Cooperate
- Afterwards: copy opponent's previous move
- Simple, retaliatory, and forgiving

Tit-for-Two-Tats (TF2T)

- Defects only after two consecutive opponent defections
- More tolerant to occasional mistakes

Grim Trigger (GT)

- Cooperate until opponent defects once
- After that: defect forever

Random (RAND)

- Chooses C or D randomly
- Represents noisy or unpredictable agents

Generous Tit-for-Tat (GTFT)

- Similar to TFT but occasionally forgives defection
- Helps maintain cooperation in noisy environments

Suspicious Tit-for-Tat (STFT)

- Starts with defection
- Then copies opponent's last move

Pavlov / Win-Stay Lose-Shift

- Repeat previous move if payoff was good
- Switch action if payoff was bad

7.2 Iterated Interaction

Each pair of connected nodes in the network (agents) plays multiple rounds (r) in a generation, and the total payoff is the sum of all the rounds.

7.3 Network Types

Agents interact over a network, refer to the class lecture, codes for more details

- Fully connected
- Random (Erdős–Rényi)
- Small-world (Watts–Strogatz)
- Scale-free (Barabási–Albert)

7.4 Evolution Across Generations

After each generation:

- Compute payoff P_i
- Reproduce strategies proportionally

$$P(\text{strategy } i) = \frac{P_i}{\sum_j P_j}$$

7.5 Linking to selected Real Data observations made in individual assignments submitted earlier

You may think of a strategy mixture to explain your observational probabilities and their transitions:

$$\theta = (\theta_1, \theta_2, ..., \theta_K)$$

$$\sum \theta_k = 1$$

Minimize:

$$\theta^* = \arg\min_{\theta} \sum_{i,j} (P_{data} - P_{sim})^2$$

Error Metric, such as the below can be used
KL Divergence:

$$D_{KL}(P_{data} || P_{sim})$$